EXACT QUOTIENT-MULTIPLICITY EXCLUSION OF 18-ELEMENT EQUALITY CANDIDATES IN THOMPSON'S GROUP F

ARGUS OPEN RESEARCH PROJECT

Abstract. In the investigation of finite Følner/amenability equality candidates in Thompson's group F on the ring-shifted $p = 22$ template, we prove that all 18-element equality candidates are impossible: the ordered-grid balance requires adjacent rows to share at least 15 right-quotient products, whereas exhaustive exact evaluation of the quotient table proves that the maximum negative-quotient multiplicity is 8.

1. Main Exclusion Theorem

Theorem 1.1. *In the $p = 22$ two-column derived template inside Thompson's group F, any 18-element configuration satisfying the boundary-cancellation equality between generators x_0 and x_1 requires adjacent row fibres to share at least 15 common right quotients, whereas the complete exact quotient table has maximum off-diagonal quotient multiplicity $8 < 15$. Hence no 18-element equality witness exists in this family.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001578 in the domain of geometric group theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped family exclusion* deciding the scope: maximum negative-quotient multiplicity $8 < 15$ excludes all 18-element equality candidates on the p=22 ring-shifted template.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- amenability of Thompson's group F.

References

[1] Primary problem list and source literature for Result 20001578 (geometric group theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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